

Seth Temple

STATISTICIAN AND APPLIED SCIENTIST

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Education

University of Washington, PhD, Statistics	2024
University of Washington, MS, Statistics	2022
University of Oregon, BS, Mathematics (<i>summa cum laude</i>)	2018

Experience

University of Michigan | Postdoctoral Researcher 09.2024 – Present

- Proposed infection rate estimators in epidemics with partial case data, & compared performance versus Bayesian approaches with data augmentation
- Built an AI models workshop and repository for postdocs, incl. sparse classifiers with CNNs and vision transformers for 100s GB of imbalanced acoustics and LSTMs to model hydrological time series

University of Washington | Graduate Student Researcher 09.2019 – 08.2024

- Created a suite of 4 methods to pinpoint and model the effects of advantageous genetic variants
- Proposed a hypothesis testing method to protect against false discoveries in case-control studies
- Devised and proved a series of new central limit theorems in population genetics
- Performed 9 whole-genome sequence analyses across 3 biobanks involving TBs of data
- Developed a fast and scalable algorithm to simulate data for biobank-scale inferences

Fred Hutch Cancer Research Center | Graduate Student Researcher 06.2023 – 12.2023

- Designed a new method to perform time series clustering for evolving SARS-CoV-2 variants
- Collaborated on an international project modeling genomic patterns of synonymous mutations

Los Alamos National Laboratory | Graduate Student Researcher 06.2020 – 10.2020

- Developed and deployed a Bayesian spatial model for mosquito-borne epidemiology
- Implemented spatial ML method to predict and visualize continent-wide mosquito prevalence

Liberty Mutual Insurance | Actuarial Assistant 07.2018 – 08.2019

- Conducted and visualized territory adjustments to ratemaking based on weather and climate risks
- Modernized SQL queries, automated quarterly reserving updates, and analyzed surety claims

University of Oregon | Natural Language Processing Researcher 01.2018 – 06.2018

- Built recurrent neural networks to predict stop words in the audio sequences of ESL students

Skills

- Python (8+ years), R (8+ years), and SQL programming
- High-performance computing (slurm, parallelizing code)
- Statistical genetics, genomics, and epidemiology
- Bayesian inference and hierarchical models
- Spatial data science and time series
- Deep learning with PyTorch and transformers
- Science communication and collaboration
- Algorithms and software development
- Bioinformatics pipelines
- Technical writing
- Project planning
- Statistical consulting

Software

- github.com/sdtemple/isweep: Python package and 5 automated pipelines for genetic analyses (4+ years of continuous software development)
- github.com/sdtemple/btvoccu: R package for Bayesian logistic regression in epidemiology
- github.com/sdtemple/peirrs & github.com/sdtemple/pblas: R packages for estimators in epidemics
- github.com/sdtemple/how-to-use-ai: tutorials for using AI coding assistants
- github.com/sdtemple/your-first-model: hack-a-thon setup for deep learning on acoustics data

Awards

- Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship [UM]
- National Defense Science and Engineering Graduate Fellowship (2% selection rate) [UW]
- Z.W. Birnbaum award for best PhD candidate exam [UW]
- NIH Traineeship in Statistical Genetics [UW]
- Phi Beta Kappa academic honor society [UO]

Teaching

- Lead Instructor for Graduate Statistics Course in Statistical Genetics [UW]
- Teaching Assistant: Intro to Machine Learning, Experimental Design, Applied Regression [UW]
- Directed Reading Program Mentor for 4 students [UW]
- College Math Tutor for 4 years [UO] and K-12 Math Tutor for 1 year [Seattle Public Libraries]

Publications

6 first-author and 5 co-author publications on statistical methods in biological sciences,
2 of which are in the 8th highest cited Genetics and Genomics journal

1. Temple, S.D., & S.R. Browning. Multiple-testing corrections in selection scans using identity-by-descent segments. *The American Journal of Human Genetics* (2025).
2. Temple, S.D., Waples, R.K., & S.R. Browning. Modeling recent positive selection using identity-by-descent segments. *The American Journal of Human Genetics* (2024).
3. Temple, S.D., & Thompson, E.A. Identity-by-descent segments in large samples. *Theoretical Population Biology* (2025).
4. Temple, S.D., Browning, S.R., & Thompson, E.A. Fast simulation of identity-by-descent segments. *Bulletin of Mathematical Biology* (2025).
5. Temple S.D., et. al. Multiple-testing corrections in case-control studies using identity-by-descent segments. *bioRxiv* (2025). [Under review at *The American Journal of Human Genetics*]
6. Temple, S.D., Manore, C.A., & Kaufeld, K.A. Bayesian time-varying occupancy model for West Nile virus in Ontario, Canada. *Stochastic Environmental Research and Risk Assessment* (2022).
7. Temple, S.D. Statistical Inference Using Identity-by-Descent Segments: Perspectives on Recent Positive Selection. University of Washington (Ph.D. thesis, 2024).
8. Temple, S.D. The Tweedie Index Parameter and Its Estimator: An Introduction with Applications to Actuarial Ratemaking. University of Oregon (Bachelor of Science thesis, 2018).
9. Haddox, H.K., Angehrn, G., Sesta, L., Jennings-Shaffer, C., Temple, S.D., et.al. SARS-CoV-2's mutation rate is highly variable between sites and is influenced by sequence context, genomic region, and RNA structure. *Nucleic Acids Research* (2025).
10. Gorris, M.E., Bartlow, A.W., Temple, S.D., et. al. Updated distribution maps of predominant *Culex* mosquitoes across the Americas. *Parasites and Vectors* (2021).
11. Browning, S.R., Temple, S.D., & Browning, B.L. FLARE2: local ancestry inference with poorly-matched reference panels. *bioRxiv* (2025). [Under review at *PLoS Genetics*]
12. Horimoto, A.R.V.R., et. al. Admixture mapping implicates 13q33.3 as ancestry-of-origin locus for Alzheimer disease in Hispanic and Latino populations. *HGG Advances* (2023).
13. Haddox, H.K., et. al. Clonal interference and changing selective pressures shape the escape of SARS-CoV-2 from hundreds of antibodies. *bioRxiv* (2025). [Under review]